

Queue Arch

SRC → Source Queue



packet - 0
drop info - 1
commands
other info

Sniffer Message

- magic number : int 32?
- type : int 16?
- interface : byte?
- length : int 64?
- magic number : int 32?

logic:

Source Queue is notified of a new buffer (via IRQ)

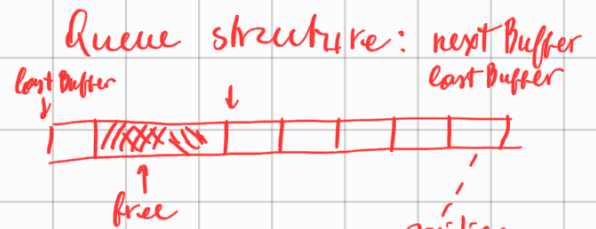
Source Queue checks that it has enough space

Source Queue updates watermarks and pointers

Source Queue initiates DMA copy of buffer

Source Queue waits for DMA copy to finish

Source Queue marks that a new buffer is ready



position
length

• next Pos =
last Buffer
pos + length

• watermark =
= next Buffer . pos

Queue Manager iterates over sources, trying to find one with buffers

Queue Manager finds a source with a buffer ready

Queue Manager identifies destinations for the source buffer
for each Destination Queue

Queue Manager checks that it has enough space

Queue Manager updates watermarks and pointers

Queue Manager initiates DMA copy of buffer

Queue Manager waits for all DMA copies to finish

once a copy is finished, triggers "send" on Destination

Queue Manager updates Source Queue watermarks and pointers,
effectively "freeing" space consumed by the buffer

Destination Queue is notified of a new buffer (by Queue Manager)

Destination Queue initiates send of a buffer via DMA to Dest.

Destination Queue waits for DMA copy to finish

Destination Queue updates its pointers and watermark, "freeing" space previously consumed by the buffer